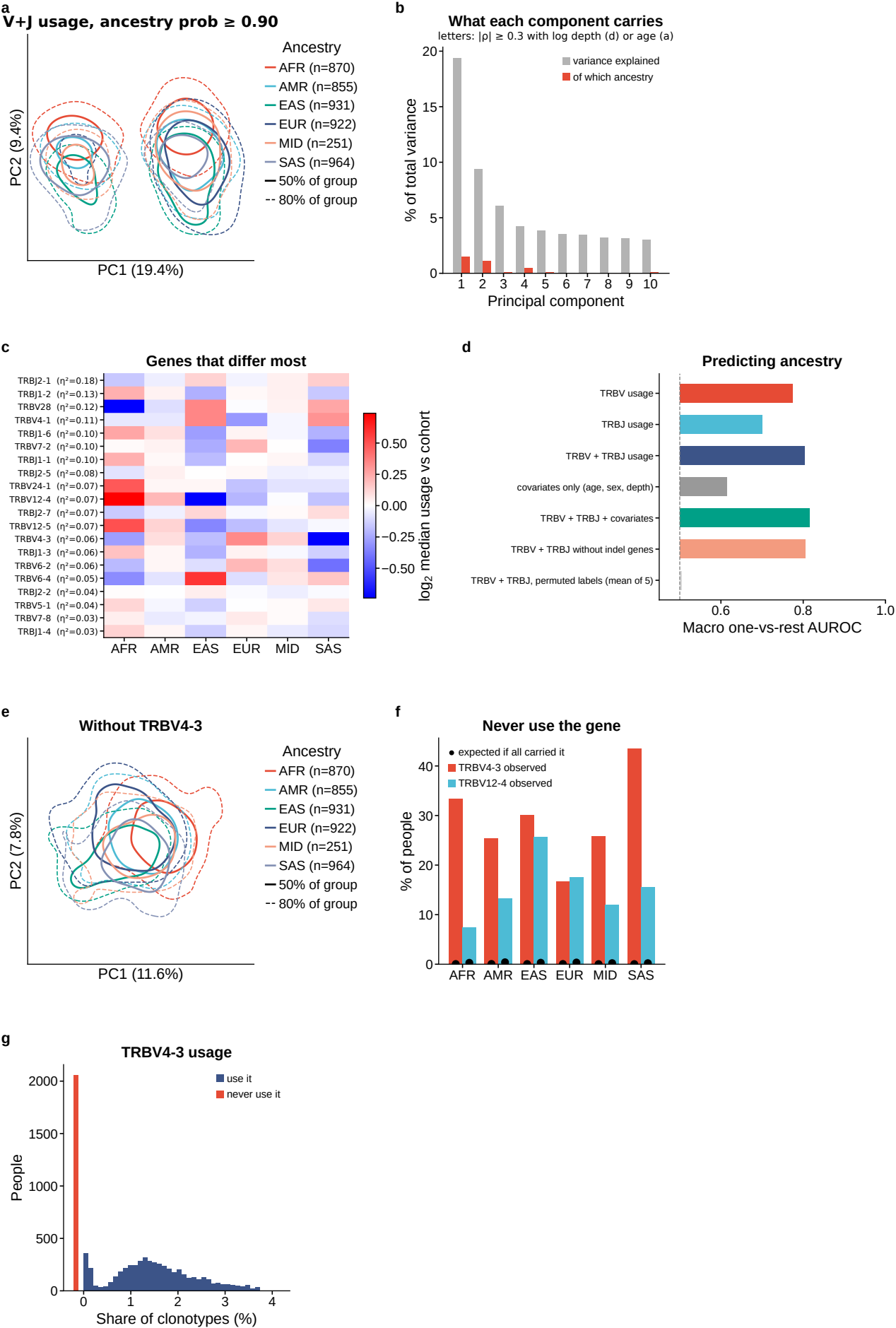

a
V+J usage, ancestry prob ≥ 0.90
PC2 (9.4%)
PC1 (19.4%)
Ancestry
AFR (n=870)
AMR (n=855)
EAS (n=931)
EUR (n=922)
MID (n=251)
SAS (n=964)
50% of group
80% of group
b
What each component carries
letters: |ρ| ≥ 0.3 with log depth (d) or age (a)
variance explained
of which ancestry
% of total variance
Principal component
c
Genes that differ most
TRBJ2-1 (η²=0.18)
TRBJ1-2 (η²=0.13)
TRBV28 (η²=0.12)
TRBV4-1 (η²=0.11)
TRBJ1-6 (η²=0.10)
TRBV7-2 (η²=0.10)
TRBJ1-1 (η²=0.10)
TRBJ2-5 (η²=0.08)
TRBV24-1 (η²=0.07)
TRBV12-4 (η²=0.07)
TRBJ2-7 (η²=0.07)
TRBV12-5 (η²=0.07)
TRBV4-3 (η²=0.06)
TRBJ1-3 (η²=0.06)
TRBV6-2 (η²=0.06)
TRBV6-4 (η²=0.05)
TRBJ2-2 (η²=0.04)
TRBV5-1 (η²=0.04)
TRBV7-8 (η²=0.03)
TRBJ1-4 (η²=0.03)
AFR AMR EAS EUR MID SAS
log2 median usage vs cohort
d
Predicting ancestry
TRBV usage
TRBJ usage
TRBV + TRBJ usage
covariates only (age, sex, depth)
TRBV + TRBJ + covariates
TRBV + TRBJ without indel genes
TRBV + TRBJ, permuted labels (mean of 5)
Macro one-vs-rest AUROC
e
Without TRBV4-3
PC2 (7.8%)
PC1 (11.6%)
Ancestry
AFR (n=870)
AMR (n=855)
EAS (n=931)
EUR (n=922)
MID (n=251)
SAS (n=964)
50% of group
80% of group
f
Never use the gene
expected if all carried it
TRBV4-3 observed
TRBV12-4 observed
% of people
AFR AMR EAS EUR MID SAS
g
TRBV4-3 usage
use it
never use it
People
Share of clonotypes (%)